When Can You Trust an Equilibrium Counterfactual?

A certified identification trichotomy for interventions on learning systems

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Abstract

Equilibrium models answer interventional queries by mutilate-and-re-solve: impose the intervention on the structural equations and read the new fixed point — the do-semantics of cyclic structural causal models. Learning and agent-based models answer the same query by unrolling adaptive dynamics and observing where they go. The two operators can disagree, and the disagreement is the formal content of a fault line running from the Lucas critique to modern complexity economics. We make the question — *when do the two operators return the same answer?* — a machine-checkable identification problem. Our main theorem is a trichotomy for global interventional queries on populations of adaptive learners: the equilibrium counterfactual is *point-identified* exactly under an expectational-stability condition (E-stability $\Leftrightarrow$ causal validity of the σ-solution, with the sharp learning-rate threshold); it is *set-identified* by a linear program over the ε-coarse-correlated-equilibrium polytope of the intervened game, at the regret ε_T actually measured at the horizon actually run, with no further assumptions; and otherwise it is *not identified*, but the failure itself is certifiable, with diagnostics — including the case of multiple stable solutions, where the honest global object is a basin-mass mixture that no plain SCM can represent. Every condition in the trichotomy is computable by instruments we release as an open-source artifact, and each returns a typed certificate. Four experiments exercise every rung: a cobweb ladder where the certificate fires correctly across four regimes, a game with non-convergent learning where the certified interval catches what the Nash point misses, a monetary-policy toy where the *sign* of a policy conclusion flips between the two semantics exactly where the certificate says not to trust the equilibrium, and a bistable system where locally correct certificates coexist with a measured global selection gap.

1 Introduction

Ask a New-Keynesian toy model what happens to inflation after a unit demand shock under a passive interest-rate rule, and the equilibrium answer is a *fall* of 8.2 points: mutilate the structural equations, re-solve the fixed point, read off the counterfactual. Ask the same structural equations populated by adaptive learners, and inflation *rises without bound* — +250 points within thirty learning steps (Section 6.3). Same model, same intervention, same query; opposite sign. Under an active rule, the two answers agree to machine precision. Nothing in the equilibrium computation warns which case one is in.

The pattern is old. The Lucas critique (Lucas, 1976) is an argument that interventional predictions computed from an equilibrium description can fail when the underlying adaptation re-equilibrates; the adaptive-learning program in macroeconomics (Marcet and Sargent, 1989; Evans

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and Honkapohja, 2001) showed that stability of learning — E-stability — governs whether rational-expectations equilibria are attainable at all; Dogra (2024) argues that the causal interpretation of equilibrium economics is paradox-ridden precisely because equilibrium models suppress the adjustment process; and the complexity-economics literature has long insisted that "equations" (equilibrium conditions) and "maps" (adaptive processes) are different objects (Brock and Hommes, 1997; Pangallo et al., 2019). On the causal-inference side, cyclic structural causal models give equilibrium counterfactuals an exact semantics (Bongers et al., 2021), the ODE-to-SCM construction says when an SCM is an abstraction of an underlying dynamical system (Mooij et al., 2013; Rubenstein et al., 2017), and Blom et al. (2019) prove that when equilibria depend on initial conditions, no plain SCM can represent the system at all. What none of these literatures provides is an *identification theory*: a statement of exactly when the equilibrium do-operator answers the interventional question posed of the adaptive system, what can be salvaged when it does not, and a procedure that checks the conditions and says so.

This paper provides that theory for two well-delimited classes — cyclic SCMs with smooth mean fields, and finite games — together with the instruments that make every condition checkable. Concretely, we contribute:

1. **A bridge theorem (Theorems 3.2 and 3.4).** For linear cyclic SCMs, point identification of the learning-limit counterfactual by the equilibrium counterfactual holds *if and only if* the intervened system is E-stable in the sense of Evans and Honkapohja (2001) — equivalently, iff the mean-dynamics Jacobian is Hurwitz — with the sharp constant-gain threshold γ^* in closed form. The theorem turns "E-stability" into a statement of causal semantics: *the σ-solution of the cyclic SCM is the causally correct description of the learning population exactly on the E-stable class.* A corollary formalizes the equations-vs-maps gap: naive unrolling demands the strictly stronger condition $\rho(B) < 1$, and the regime in between — divergent map, convergent learners — is nonempty and certifiable. Theorem 3.4 extends both directions to hyperbolic equilibria of smooth systems, with explicit noise conditions, at the price of a basin-local scope.

2. **An identification trichotomy (Theorem 5.2).** For global interventional queries posed of learning populations in the two classes, exactly one of three cases holds, each decidable by a finite computation and each mapped to a typed certificate: *point* identification (IDENTIFIED), *set* identification by LP bounds over the ε_T-coarse-correlated-equilibrium polytope of the intervened game at the *measured* realized regret ε_T (BOUNDED; Theorem 4.1, which is deliberately assumption-free and binds at every finite horizon), and *certified non-identification* (EMPIRICAL) with diagnostics.

3. **Certified failure rather than silent failure.** In the non-identified case the instruments report *why*: a negative stability margin with the proof that no learning rate rescues convergence; a vacuous interval, reported as abstention rather than as a bound; or multiplicity of stable solutions, where the global interventional object is a basin-mass mixture — the causal-constraint-model caveat of Blom et al. (2019) operationalized, and, in our experiments, *measured*.

4. **Instruments and evidence.** All conditions are computed by exact, dependency-free instruments we release as an open-source artifact: a stability-margin diagnostic and a damped-unrolling comparator for cyclic SCMs, and a CCE-polytope constructor with LP bounds, measured-regret certificates, and dual-multiplier sensitivities for finite games. Four experiments exercise every rung of the resulting certificate ladder (Section 6).

What this paper is not. The individual mathematical ingredients are standard — stochastic-approximation theory on one side, LP duality over equilibrium polytopes on the other — and Theorem 4.1 in particular is a rearrangement of the definition of regret. The contribution is the synthesis: the two-semantics framing, the bridge that makes E-stability a causal-validity condition, the trichotomy that makes "when can you trust an equilibrium counterfactual" a decidable question with typed answers, and the instruments plus experiments that demonstrate the ladder end to end. We state each theorem's provenance next to it, and we flag every place where our experimental claims stop short (Section 8).

2 Two do semantics for one query

2.1 Equilibrium do: cyclic SCMs

A cyclic SCM over endogenous variables $x \in \mathbb{R}^n$ and exogenous variables u specifies structural equations $x_i = f_i(x_{\mathrm{pa}(i)}, u_i)$ whose dependency graph may contain cycles. Its solutions are the fixed points $x = f(x, u)$; Bongers et al. (2021) develop the measure-theoretic solution-function (σ-solution) semantics under which such models support interventions and counterfactuals. The **equilibrium** do of an intervention is mutilate-and-re-solve: replace the targeted equations, then solve the intervened fixed-point problem. Throughout, our exact linear class is

$$x = Bx + u, \qquad \mathrm{do}(x_i{=}\xi): \text{ zero row } i \text{ of } B, \text{ pin } u_i = \xi, \tag{1}$$

with intervened pair $(B_{\mathrm{do}}, u_{\mathrm{do}})$ and, whenever $I - B_{\mathrm{do}}$ is invertible, the unique equilibrium counterfactual

$$x^* \;=\; (I - B_{\mathrm{do}})^{-1}\, \mathbb{E}[u_{\mathrm{do}}]. \tag{2}$$

We write $\mathrm{do}^{\mathrm{eq}}(\cdot)$ for this operator. For finite games (Section 4), the analogous equilibrium object is an equilibrium set of the intervened game rather than a point; the correlated-equilibrium polytope will play the role of $(I - B)^{-1}$.

2.2 Learning do: adaptive dynamics

The same structural equations, populated by agents who do not solve fixed points but adapt, define a process: impose the same mutilation on the *process* and unroll,

$$x_{k+1} \;=\; x_k + \gamma_k\big(f_{\mathrm{do}}(x_k) - x_k + M_{k+1}\big), \tag{3}$$

where γ_k is a gain sequence and M_{k+1} is martingale-difference noise. Constant gain $\gamma_k \equiv \gamma \in (0, 1]$ is the Euler discretization of the **mean dynamics**

$$\dot{x} \;=\; F_{\mathrm{do}}(x) \;\coloneqq\; f_{\mathrm{do}}(x) - x, \tag{4}$$

and decreasing gain ($\sum_k \gamma_k = \infty$, $\sum_k \gamma_k^2 < \infty$) is Robbins–Monro stochastic approximation (Robbins and Monro, 1951) — the regime of least-squares learning in macroeconomics (Marcet and Sargent, 1989; Evans and Honkapohja, 2001). Equation (4) is the E-stability ODE of Evans and Honkapohja (2001) in mean-dynamics form: f_{do} is the temporary-equilibrium map from expectations to realized outcomes, and E-stability of an equilibrium x^* is local asymptotic stability of x^* under (4). The **learning** do, written $\mathrm{do}^{\mathrm{lr}}(\cdot)$, is the long-run behavior of (3): its almost-sure limit when one exists, and its time-averaged play otherwise.

Two spectral quantities decide everything in the linear class. With $\nu_i = \lambda_i(B_{\mathrm{do}}) - 1$ the eigenvalues of the mean-dynamics Jacobian $B_{\mathrm{do}} - I$,

$$\mathrm{margin}(B_{\mathrm{do}}) \;=\; 1 - \max_i \operatorname{Re} \lambda_i(B_{\mathrm{do}}), \qquad \gamma^* \;=\; \min_i \frac{-2\operatorname{Re}\nu_i}{|\nu_i|^2}, \tag{5}$$

the *stability margin* (positive iff $B_{\mathrm{do}} - I$ is Hurwitz) and the *maximal stable learning rate* (the sharp constant-gain threshold, Lemma 3.1). Both are computed exactly by the released instruments (Appendix E).

2.3 Queries, and what "identified" means here

A *query* is the long-run population value $Q = \mathbb{E}_\mu[h]$ of a functional h under the intervention, where μ is the population's long-run occupation law. We distinguish *global* queries — the population's long-run behavior, full stop — from *basin-local* queries conditioned on initialization within the basin of a particular stable solution. The distinction is not pedantic: Section 6.4 measures a system where every basin-local query is point-identified while the global query is not. We say the equilibrium counterfactual *point-identifies* the query when $\mathrm{do}^{\mathrm{lr}}(\cdot)$ exists as a point and equals $\mathrm{do}^{\mathrm{eq}}(\cdot)$; *set-identifies* it when a computable set provably contains the realized value; and *fails* otherwise.

2.4 Certificates

Every instrument returns a typed certificate: a rung in $\{\textsc{Identified}, \textsc{Bounded}, \textsc{Empirical}\}$, a claim, a machine-checkable *witness* (spectra, LP values, dual multipliers, measured regrets), and *hedges* — explicit statements of what blocked a stronger rung and, when possible, what action would unblock it (e.g. "reduce the learning rate below γ^*"). The rung ordering is a soundness contract, not a scoring rule: IDENTIFIED may be claimed only under a proof of point identification, BOUNDED only under a proof of containment, and everything else must degrade to EMPIRICAL with diagnostics rather than fabricate a bound. The do-comparator for cyclic SCMs and the game certifier for finite games implement the ladder in Table 1.

3 Point identification: E-stability is causal validity

Lemma 3.1 (gain window)**.** *Let $\nu \in \mathbb{C}$. If $\operatorname{Re}\nu < 0$ then $|1+\gamma\nu| < 1$ iff $0 < \gamma < -2\operatorname{Re}\nu/|\nu|^2$. If $\operatorname{Re}\nu \geq 0$ then $|1+\gamma\nu| \geq 1$ for every $\gamma > 0$.*

Proof. $|1+\gamma\nu|^2 - 1 = \gamma\,(2\operatorname{Re}\nu + \gamma|\nu|^2)$, which is negative iff $\gamma < -2\operatorname{Re}\nu/|\nu|^2$, a positive threshold iff $\operatorname{Re}\nu < 0$; if $\operatorname{Re}\nu \geq 0$ the right factor is nonnegative for all $\gamma > 0$. □

Theorem 3.2 (T1: point identification ⇔ E-stability; linear class)**.** *Let $x = Bx + u$ be a linear cyclic SCM with intervened pair $(B_{\mathrm{do}}, u_{\mathrm{do}})$, $I - B_{\mathrm{do}}$ invertible, x^* as in (2), and learning dynamics (3) with $f_{\mathrm{do}}(x) = B_{\mathrm{do}}x + u_{\mathrm{do},k}$, $\mathbb{E}[u_{\mathrm{do},k}] = \mathbb{E}[u_{\mathrm{do}}]$, and noise with bounded conditional second moments. The following are equivalent:*

(1) E-stability: $\mathrm{margin}(B_{\mathrm{do}}) > 0$*, i.e. $B_{\mathrm{do}} - I$ is Hurwitz;*

(2) constant gain: *for every $\gamma \in (0, \gamma^*)$ the mean iteration converges to x^* from every initial condition, and the stochastic iterates satisfy* $\limsup_k \mathbb{E}\|x_k - x^*\|^2 = O(\gamma)$*;*

(3) decreasing gain: *Robbins–Monro learning converges to x^* almost surely (on the event that the iterates remain bounded, which under (1) has probability one for the linear recursion).*

Under any of these, $\mathrm{do}^{\mathrm{lr}}(\cdot) = \mathrm{do}^{\mathrm{eq}}(\cdot)$*: the* σ*-solution of the intervened cyclic SCM is the causally correct description of the learning population's interventional behavior. Conversely, if* $\mathrm{margin}(B_{\mathrm{do}}) < 0$*: for every* $\gamma > 0$ *the mean iteration diverges from all initial conditions outside a proper subspace, and with noise nondegenerate in the unstable directions, decreasing-gain learning converges to* x^* *with probability zero. The learning-limit* do *then does not exist as a point, and the equilibrium prediction has no dynamical counterpart.*

Proof (algebraic part; stochastic parts in Appendix A). The mean iteration is $x_{k+1} = M_\gamma x_k + \gamma \mathbb{E}[u_{\mathrm{do}}]$ with $M_\gamma = I + \gamma(B_{\mathrm{do}} - I)$, whose fixed point is x^* for *every* γ — damping never moves the equilibrium, only the convergence class. Its eigenvalues are $1 + \gamma\nu_i$; by Lemma 3.1, $\rho(M_\gamma) < 1$ for all $\gamma \in (0, \gamma^*)$ iff every $\mathrm{Re}\,\nu_i < 0$, i.e. iff (1), and γ^* in (5) is sharp. If some $\mathrm{Re}\,\nu_i > 0$ then $|1 + \gamma\nu_i| > 1$ for all $\gamma > 0$, giving divergence along that eigenspace. The noise bound in (2), the ODE-method equivalence (1)⇔(3) (Ljung, 1977; Kushner and Yin, 2003; Benaïm, 1999), and the probability-zero converse (Pemantle, 1990; Brandière and Duflo, 1996) are in Appendix A. □

Theorem 3.2 is a bridge, and we claim it as such: each direction is classical in its home literature (the spectral algebra and the ODE method in stochastic approximation; E-stability in Evans and Honkapohja, 2001), but the equivalence has, to our knowledge, never been stated as a statement about the validity of SCM do-semantics — "E-stability" becomes "the σ-solution is the causally correct object," a sentence neither literature could previously parse.

Corollary 3.3 (naive unrolling is a strictly stronger demand)**.** *The literal period-*1 *unrolling* $x_{k+1} = B_{\mathrm{do}} x_k + u$ *converges iff* $\rho(B_{\mathrm{do}}) < 1$*. Since* $\rho(B) < 1 \Rightarrow \max_i \mathrm{Re}\,\lambda_i(B) < 1$ *but not conversely, the regime* $\{\rho(B_{\mathrm{do}}) \geq 1,\ \mathrm{margin}(B_{\mathrm{do}}) > 0\}$ *is nonempty (e.g.* $B = [-2]$*:* $\rho = 2$*,* $\mathrm{margin} = 3$*): the map diverges while every sufficiently damped adaptive population converges to the equilibrium counterfactual. This is the formal content of the equations-vs-maps gap for local interventions in the linear class, and it is exactly what the comparator certifies when called with a learning rate below* γ^**.*

3.1 The nonlinear local statement

Let $F = f - \mathrm{id}$ with $f \in C^1$, and let x^* be a *hyperbolic* zero of F (no eigenvalue of $J = DF(x^*)$ on the imaginary axis). The full assumptions (A1)–(A4) — smoothness, martingale noise with conditionally bounded second moments, Robbins–Monro gains, and, for the converse, a noise-excitation condition — are stated in Appendix B.

Theorem 3.4 (T1′: hyperbolic local version)**.** *Under (A1)–(A3):*

(i) If J *is Hurwitz, then for every* $\gamma \in (0, \gamma^*(J))$ *(the formula in (5) evaluated on the spectrum of* J*) the constant-gain iteration is locally exponentially stable at* x^* *with asymptotic mean-square error* $O(\gamma)$*, and decreasing-gain iterates converge to* x^* *almost surely on the event of remaining in a compact subset of the basin of attraction of* x^**. Hence* $\mathrm{do}^{\mathrm{lr}}(\cdot) = \mathrm{do}^{\mathrm{eq}}(\cdot)$ locally, on the basin of x^*.

(ii) If J *has an eigenvalue with positive real part, then under the additional excitation condition (A4),* $\mathbb{P}(x_k \to x^*) = 0$*: the equilibrium counterfactual at* x^* *has no learning-limit counterpart.*

Proof sketch. Hyperbolicity reduces (i) to the linear case via the Hartman–Grobman linearization for the deterministic part and the limit-set theorem of stochastic approximation for the almost-sure part (Kushner and Yin, 2003; Benaïm, 1999; Borkar, 2008); (ii) is the nonconvergence-to-unstable-points theorem (Pemantle, 1990; Brandière and Duflo, 1996). Full proof with the noise conditions made explicit: Appendix B. □

Remark 3.5 (the basin qualifier binds). T1$'$ is local by nature. When the intervened system has *multiple* stable σ-solutions, every basin-local certificate can be simultaneously correct while the global interventional outcome is a *basin-mass mixture* across them — and interventions move both the solutions and the basin boundaries. Blom et al. (2019) prove that equilibria depending on initial conditions are exactly what plain SCM semantics cannot represent; equilibrium selection by learning is that phenomenon, not merely an analogy to it. Section 6.4 measures the resulting local-right/global-wrong gap. This is why the trichotomy below is stated for global queries and treats multiplicity as certified non-identification with diagnostics rather than as a larger point-identified case.

4 Set identification: the measured-regret CCE rung

The second exact class is finite games, where the equilibrium object is a set and the honest counterpart of (2) is a polytope. Let G be a finite game with agents N, action sets A_i, and utilities u_i; an intervention pins the actions of a subset of agents, inducing G_{do}: profiles restricted to the consistent ones, no-deviation constraints retained only for free agents. For $\varepsilon = (\varepsilon_i)_i$, the ε-coarse-correlated-equilibrium polytope is

$$\mathrm{CCE}_\varepsilon(G_{\mathrm{do}}) \;=\; \Big\{\mu \in \Delta(\text{profiles}) \;:\; \mathbb{E}_\mu\big[u_i(a_i', s_{-i}) - u_i(s)\big] \le \varepsilon_i \;\; \forall i \text{ free},\; \forall a_i' \in A_i\Big\}. \tag{6}$$

A population plays T rounds of G_{do}, producing the empirical joint distribution μ_T; agent i's *realized regret* is $\varepsilon_T(i) = \max_{a_i'} \frac{1}{T}\sum_{t=1}^{T}\big[\,u_i(a_i', s_{-i,t}) - u_i(s_t)\,\big]$ clipped at zero — computable from the trajectory log alone.

Theorem 4.1 (T2: finite-time containment, assumption-free)**.** *For every horizon T and every profile functional h,*

$$\mathbb{E}_{\mu_T}[h] \;\in\; \Big[\min_\mu \mathbb{E}_\mu[h],\; \max_\mu \mathbb{E}_\mu[h]\Big] \quad \textit{over } \mu \in \mathrm{CCE}_{\varepsilon_T}(G_{\mathrm{do}}),$$

where ε_T is the vector of measured realized regrets. Both endpoints are linear programs over a polytope contained in the simplex.

Proof. $\mu_T \in \mathrm{CCE}_{\varepsilon_T}(G_{\mathrm{do}})$ is the definition of realized regret rearranged: the expected deviation gain of switching to a fixed a_i' under μ_T *is* the time-averaged realized gain, which is at most $\varepsilon_T(i)$ by construction. Containment of $\mathbb{E}_{\mu_T}[h]$ between the min and max over the feasible set is then immediate. □

Theorem 4.1 is definitionally true, and we present it as such: its role in this paper is to be the *set-identification rung of the trichotomy*, in the certified form that binds at the horizon actually run, for whatever the learners actually did, abstaining when vacuous. It is horizon-uniform — it holds at every T simultaneously because each instance is deterministic given the log — which we prefer to the term "anytime-valid" and its martingale connotations. Its development into a full empirical instrument (sensitivity analysis at scale, and the demarcation of which learning families respect the stage-game polytope at long horizons) is companion work; here we record the three corollaries the trichotomy needs.

Corollary 4.2 (asymptotic route)**.** *If the population is no-regret ($\varepsilon_T \to 0$; Foster and Vohra, 1997; Hart and Mas-Colell, 2000), every limit point of $\{\mu_T\}$ lies in $\mathrm{CCE}_0(G_{\mathrm{do}})$, and since the LP value is a Lipschitz function of the constraint right-hand side (with constant given by the optimal dual multipliers), the ε_T-bounds converge to the exact-CCE bounds. The classical statement is the $\varepsilon \to 0$ limit of the certified one.*

Corollary 4.3 (degeneracy: when the equilibrium point prediction is safe)**.** *The equilibrium point prediction of h is valid for every no-regret population iff h is constant over* $\mathrm{CCE}_0(G_{\mathrm{do}})$ *(LP width* 0*) — a checkable condition, certified* IDENTIFIED*. Example: in two-player zero-sum games every CCE achieves the value, so value functionals are point-identified for any no-regret population; generic anti-coordination games are not.*

Corollary 4.4 (ε-sensitivity from LP duals)**.** *Each endpoint of the interval is a concave (max side) or convex (min side) piecewise-linear function of ε, and its one-sided derivative at the measured ε_T is the sum of the optimal dual multipliers on the perturbed constraints — returned by the solver and reported in the certificate witness. Reading: multiplied by the population's regret decay rate, it prices the marginal certificate tightness bought by running the population longer; an interval that is wide while its sensitivity is tiny is* structurally *loose (the game), not statistically loose (the learners) — a distinction no asymptotic statement makes.*

Remark 4.5 (width is the operational content of "equilibrium")*.* $\mathrm{width}(h, \mathrm{do}) = \max - \min$ over $\mathrm{CCE}_0(G_{\mathrm{do}})$ measures how much predictive content equilibrium analysis has for adaptive populations under that intervention: zero is full point identification; the full payoff range is none, and the certificate then *abstains* rather than reporting a vacuous interval.

Remark 4.6 (scope: the stage-game polytope)*.* $\mathrm{CCE}_\varepsilon(G_{\mathrm{do}})$ is the *stage-game* polytope. Populations with history-dependent strategies and intertemporal objectives — the folk-theorem ingredients, and empirically the value-bootstrapping learners of the algorithmic-collusion literature (Calvano et al., 2020) — can sustain time-averages outside it. Theorem 4.1 is not violated: such populations are *detected, not missed*, because their measured ε_T stays bounded away from zero and the certificate inflates or abstains instead of endorsing the stage-game analysis. Mapping which learning families escape, and reading the measured ε_T as a diagnostic of what enforcement is forgoing, is the subject of the companion paper; the trichotomy below only needs that the containment claim itself is unconditional.

Provenance. Every atomic ingredient of this section exists in some form: set-valued counterfactuals over Bayes-correlated-equilibrium polytopes (Bergemann et al., 2022); LP identified sets over equilibrium polytopes with asymptotic no-regret justifications (Magnolfi and Roncoroni, 2023; Syrgkanis et al., 2017); no-regret as the identifying behavioral assumption (Nekipelov et al., 2015); set-valued interventions over equilibrium sets in causal games (Hammond et al., 2023; Mishra and Fox, 2024); and the LP-over-correlated-equilibria machinery itself (Papadimitriou and Roughgarden, 2008). The delta claimed here is exactly: the finite-time *measured-regret certified* form with abstention; the trichotomy stated against the cyclic-SCM do as two semantics for one query; and width as the operational equilibrium-content measure. The LP lives in joint-profile space (exponential in the number of players; hardness results exist for succinct games, Papadimitriou and Roughgarden, 2008), so exact bounds are for small games; sampled and relaxed bounds are the scale path.

5 The main theorem: an identification trichotomy

We now assemble the pieces. The theorem classifies *global* queries (Section 2.3) for systems in one of the stated classes; its honesty hinges on two scoping decisions motivated by Remark 3.5: point identification in the SCM class requires *global* uniqueness of the stable σ-solution, and everything outside the stated regularity is routed to the diagnostics case rather than silently assumed away.

Assumption 5.1 (regularity for the nonlinear SCM class)**.** The intervened mean field $F_{\mathrm{do}} = f_{\mathrm{do}} - \mathrm{id}$ is C^1 with (R1) finitely many zeros, all hyperbolic; (R2) SA iterates remain in a compact set almost surely (dissipativity); (R3) the chain-recurrent set of the mean flow consists of its zeros only (no cycles or chaotic sets). (R1)–(R3) hold automatically in the linear class with margin $\neq 0$, and (R3) is automatic for scalar systems with hyperbolic zeros. (R2)–(R3) are not decidable a priori in general; the instruments guard them at runtime with trajectory diagnostics (divergence detection and a largest-Lyapunov estimate), and their failure routes to case 3 below.

Theorem 5.2 (identification trichotomy for learning populations)**.** *Let $Q = \mathbb{E}_\mu[h]$ be a global interventional query posed of a population of adaptive learners* (3)*, where the intervened system is (a) a linear cyclic SCM, (b) a smooth mean field satisfying Assumption 5.1, or (c) a finite game played over T rounds with measured realized regrets ε_T. Then exactly one of the following holds, each condition computable by a shipped instrument, and the corresponding certificate is warranted:*

1. ***Point (*****Identified*****).*** *Either (i) the intervened system* (a)/(b) *has a* globally unique *stable σ-solution x^* with* margin > 0 *(Hurwitz Jacobian): then for every $\gamma \in (0, \gamma^*)$, and almost surely under decreasing gain, the learning* do *equals the equilibrium* do *at x^* (Theorems 3.2 and 3.4); or (ii) the system is a finite game and h is constant over* $\mathrm{CCE}_0(G_{\mathrm{do}})$ *(LP width $\leq$ tol): the equilibrium point prediction holds for every no-regret population (Corollary 4.3).*

2. ***Set (*****Bounded*****).*** *The system is a finite game with LP width $>$ tol and a nonvacuous interval at the measured ε_T: the realized time-average of h lies in the LP interval over* $\mathrm{CCE}_{\varepsilon_T}(G_{\mathrm{do}})$*, with no further assumptions (Theorem 4.1), and the reported dual multipliers price the interval's growth per unit of additional regret (Corollary 4.4).*

3. ***None (*****Empirical,** ***hedged with diagnostics*****).*** *One of: every σ-solution has unstable mean dynamics (*margin < 0 *at each; no admissible gain rescues — Theorem 3.2 converse); Assumption 5.1 fails at runtime (divergence, cycles, or a positive Lyapunov estimate); the ε_T-interval is vacuous (abstention; Remark 4.5); or the intervened system has* multiple *stable σ-solutions, so the global learning outcome is a basin-mass mixture over them — an object plain SCM semantics cannot represent (Blom et al., 2019) — reported as a selection hedge. In the multiplicity case each stable σ-solution still carries a correct* basin-local *case-1 certificate (T1′ is local by nature); the trichotomy classifies the global query.*

Proof sketch (details in Appendix D). Within classes (a)/(b), enumerate the σ-solutions (finitely many by (R1); spectral computation in (a)) and count the stable ones: zero stable (or (R) failing) routes to case 3; exactly one stable with margin > 0 routes to case 1(i) via Theorem 3.2 (global convergence is automatic in the linear class; in class (b) it follows from (R2)–(R3) plus the limit-set theorem and nonconvergence-to-unstable-points, Appendix D); two or more stable routes to case 3 (multiplicity). Within class (c): width $\leq$ tol is case 1(ii); otherwise a nonvacuous measured-ε_T interval is case 2 and a vacuous one is case 3. The cases are exhaustive and mutually exclusive by construction — in particular 1(i) and 3(multiplicity) cannot co-fire because 1(i) requires global uniqueness, and 1(ii) and 2 cannot co-fire because they split on width. Each condition is a finite computation given the enumerated solutions: margin signs, solution count, LP width, LP vacuity (one instrument per row of Table 1). $\square$

Certified failure, not silent failure. Case 3 is where most of the practical value lies, because it is the case existing practice silently mislabels. The instruments report: with margin < 0, the geometric divergence rate along the unstable eigenspace and a proof that no learning rate rescues it; with $\rho(B_{\mathrm{do}}) \geq 1$ but margin > 0, the actionable hedge that converts an apparent divergence into a certified re-run (Corollary 3.3); with attracting cycles or chaos, a largest-Lyapunov diagnostic and

condition	instrument	certificate
$\rho(B_{\text{do}}) < 1$, gap $\leq$ tol	comparator (naive unrolling)	IDENTIFIED
margin > 0, $\gamma < \gamma^*$, gap $\leq$ tol	comparator (damped, T1)	IDENTIFIED
margin > 0 but naive unrolling diverges	comparator hedge + γ^*	EMPIRICAL (actionable)
h constant over $\text{CCE}_0(G_{\text{do}})$	game certifier (T2 degeneracy)	IDENTIFIED
no-regret assumed or ε_T measured	game certifier (T2)	BOUNDED
ε_T-interval vacuous	game certifier	EMPIRICAL (abstention)
margin < 0 at every σ-solution	comparator hedge (no rescuing γ)	EMPIRICAL (certified failure)
$I - B_{\text{do}}$ singular / margin ≈ 0 / multiplicity	typed refusal + selection hedge	EMPIRICAL (selection / CCM caveat)

Table 1: The certificate ladder as shipped: every rung of Theorem 5.2, the instrument that checks it, and the certificate it warrants. "Actionable" hedges name the re-run that would upgrade the rung (e.g. a learning rate below γ^*); the last row is a typed refusal to fabricate a solution when the equilibrium object is ill-posed or selection-dependent.

regime	certificate	key witness values
A: $s/d = 0.5$, naive	IDENTIFIED	gap 8.9×10^{-16}; $p^* = 6.3333$
B: $s/d = 1.5$, naive	EMPIRICAL (actionable)	$\rho = 1.225$ diverges; margin $= +1.0$; hedge $\gamma^* = 0.8$
B: $s/d = 1.5$, $\gamma = 0.4$	IDENTIFIED	gap 0; $p^* = 4.2000$
C: trend $b = 1.2$	EMPIRICAL (certified failure)	margin $= -0.095$; $\gamma^* = 0$: no gain rescues
D: nonlinear supply, gain 0.14	EMPIRICAL (hedge: $\gamma > \gamma^* = 0.073$)	Lyapunov $+0.222$; $f'(p^*) = -26.3$

Table 2: E1: one market, four regimes, the full ladder. Regime B is Corollary 3.3 live: the naive cobweb oscillation diverges while the margin is positive, the hedge names γ^*, and the damped re-run certifies IDENTIFIED — Nerlove's adaptive-expectations rescue, certified. Regime C is the certified-failure rung. Regime D is chaotic (positive Lyapunov exponent).

the fallback that the *time-averaged* play of a finite-game population still satisfies Theorem 4.1 at its measured regrets; with multiplicity or a singular intervened system, a typed refusal carrying all stable solutions and — when trajectories are available — ensemble basin masses (Section 6.4). Dash (2005) is the causal-side precursor: equilibration changes causal structure, so manipulating the equilibrated model misleads; the ladder is that warning made operational.

6 Experiments

Four experiments exercise every rung of Table 1. All numbers below are outputs of the released scripts with fixed seeds (parameters in Appendix E); every LP value is verified by an independent weak-duality certificate (dual-feasible multipliers with slack $\sim 10^{-14}$), so no headline number depends on trusting our simplex implementation.

6.1 E1: the certificate ladder on the canonical stage

The cobweb market — demand $D(p) = a - dp$, supply $S(p^e) = c + sp^e$ under naive expectations, per-unit tax do(τ=1) as the intervention — is the oldest stage for equilibrium-vs-dynamics questions (Nerlove, 1958; Hommes, 1994). Table 2 runs one market through four regimes.

Two honesty notes. First, regime D's chaos enters through the adaptive gain itself: with naive expectations a monotone cobweb map admits only fixed points and 2-cycles, so the device that rescues regime B is the one that produces chaos in the nonlinear class (Hommes, 1994) — the ladder's two EMPIRICAL hedges are two ends of the same instrument. Second, in regime D the

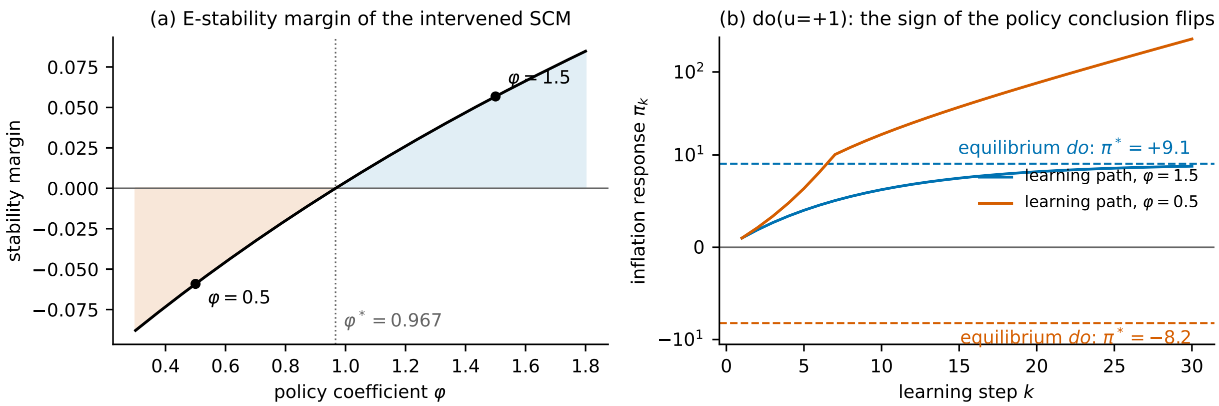

Figure 1: E4. Left: stability margin of the intervened SCM as a function of the policy coefficient φ, crossing zero at $\varphi^* = 0.9667$. Right: under do(u=+1), naive-expectations learning paths against the equilibrium predictions (dashed): at $\varphi = 1.5$ they agree (IDENTIFIED, gap 7×10^{-15}); at $\varphi = 0.5$ the equilibrium predicts $\pi^* = -8.21$ while learning explodes upward ($\pi_{30} = +249.6$) — the sign of the policy conclusion flips, and the certificate reports margin $= -0.059$ with $\gamma^* = 0$: no learning rate rescues it.

time-averaged price (5.8747) happens to sit near p^* (5.9562) because the chaotic attractor is roughly symmetric around the fixed point; the pointwise prediction fails completely (the trajectory never settles), and the certificate hedges exactly there.

6.2 E2: non-convergent learning and the Bounded rung

The perturbed rock–paper–scissors of Sato et al. (2002) (tie payoffs $\epsilon_X = 0.5$, $\epsilon_Y = -0.1$ breaking the zero-sum structure) is the canonical game in which learning dynamics fail to converge to Nash. We intervene by taxing X's first action ($\tau = 0.3$), build G_{do} explicitly, and run a Hedge population for $T = 200{,}000$ rounds. The Nash point prediction of X's payoff in G_{do} is 0.0667; the realized time-average is 0.1034 — a miss of 0.037. The measured realized regret is $\varepsilon_T = 0.0021$, and the measured-ε interval $[0.0646,\ 0.3120]$ contains the realized value, as Theorem 4.1 guarantees by construction. Two readings, both honest: the certificate catches what the Nash point misses; and the interval width (0.24 against a payoff range of ≈ 2.3) *is* the finding — it is the measured amount of predictive content that "equilibrium" has for adaptive play in this game (Remark 4.5), and the miss it absorbs is small relative to it. We label the regime "non-convergent learning" rather than chaos: Sato et al. (2002) prove chaos for replicator families, and we did not separately verify a positive Lyapunov exponent for this discrete Hedge configuration.

6.3 E4: the certified policy sign flip

A New-Keynesian-style toy in the spirit of Dogra (2024) reduces to the expectations recursion $\pi = T(\varphi)\,\pi^e + u'$ with $T(\varphi) = (\beta + \kappa\sigma)/(1 + \kappa\sigma\varphi)$, closed by naive expectations into the cyclic SCM $B = \left[\begin{smallmatrix} 0 & T \\ 1 & 0 \end{smallmatrix}\right]$ ($\beta = 0.99$, $\sigma = 1$, $\kappa = 0.3$). E-stability holds iff $T(\varphi) < 1$, i.e. $\varphi > \varphi^* = 1 - (1-\beta)/(\kappa\sigma) = 0.9667$ — a Taylor-principle-*type* threshold of this bespoke static-timing toy (it recovers the textbook $\varphi > 1$ exactly in the patient limit $\beta \to 1$; it is *not* the condition of Bullard and Mitra, 2002, whose model timing and expectations differ). Figure 1 shows the exhibit: under do(u=+1) the margin crosses zero at φ^* (-0.0059 at $\varphi^* - 0.05$, $+0.0058$ at $\varphi^* + 0.05$), and on the passive side the sign of

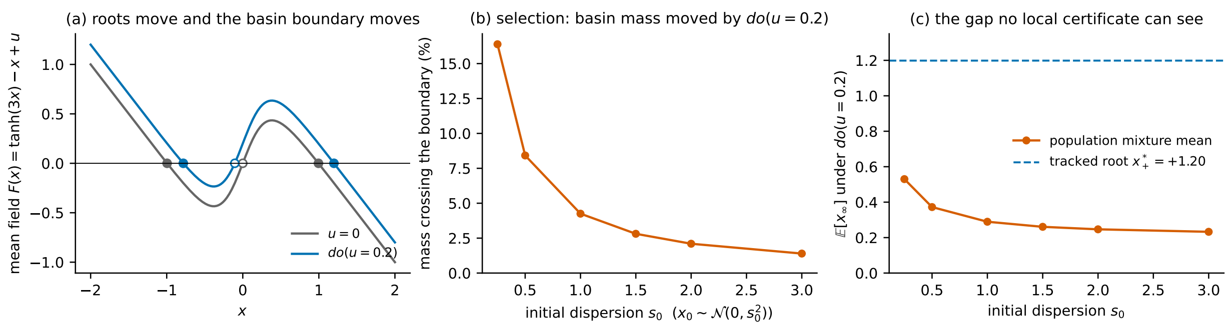

Figure 2: E7. (a) Bistable mean field $F(x) = \tanh(3x) - x + u$: under do(u=0.2) both stable roots move *and* the basin boundary moves ($0 \to -0.105$). (b) Fraction of a stochastic-approximation ensemble ($N = 10^5$, $x_0 \sim \mathcal{N}(0, s_0^2)$) crossing the basin boundary under the intervention, as a function of initial dispersion s_0. (c) The population mixture mean under do(u=0.2) against the equilibrium-tracking prediction $x_+^* = +1.199$: the gap persists at every dispersion (0.67–0.97) and is largest exactly when the population starts concentrated near the old boundary.

the policy conclusion flips between the two semantics. An economist may object that nobody trusts comparative statics at an E-unstable equilibrium; the reply, following Dogra (2024) and Evans and Ramey (2006), is that stability is routinely *not checked* — the contribution is the machine check, not the phenomenon. A standard-timing calibrated New-Keynesian learning model is deferred to the economics-facing companion paper.

6.4 E7: locally certified, globally selected

The bistable mean field $F(x) = \tanh(3x) - x + u$ has two stable equilibria and one unstable one. At $u = 0$ the stable roots ± 0.995 carry margins $+0.969$ *each*: both basin-local T1′ certificates are Identified, and both are correct. Under do(u=0.2) the roots move to -0.782 and $+1.199$ (margins $+0.892$, $+0.991$) and the basin boundary moves from 0 to -0.105; a stochastic-approximation ensemble ($N = 10^5$, gain 0.05, noise 0.1, $x_0 \sim \mathcal{N}(0, 1.5^2)$) re-partitions from a 50.2/49.8 split to 47.4/52.6 — 2.8% of the population crosses the boundary. Equilibrium-tracking at the positive root predicts $+1.199$; the realized population mixture mean is $+0.261$ — a gap of 0.938 that no basin-local certificate can see (tracking the negative root instead gives 1.04; the conclusion is root-invariant). This is Theorem 5.2's case-3 multiplicity branch, and the Blom et al. (2019) caveat, *measured*.

Because the basin masses depend on the (arbitrary) initial distribution, we report the whole curve rather than one point (Figure 2b,c): as the initial dispersion s_0 ranges over 0.25–3.0, the crossing mass falls from 16.4% to 1.4%, while the gap between the tracked root and the mixture mean persists at every dispersion (0.67–0.97) and is largest when the population starts concentrated near the old boundary — the near-indifference case. The selection effect is not an artifact of the initial-condition choice; its *size* is the curve. The diagnostic this motivates — a multiplicity-aware hedge reporting all stable σ-solutions and, when trajectories are available, ensemble basin masses — is what the last row of Table 1 shows and what E7 computes explicitly.

7 Related work

Cyclic SCMs and dynamical abstraction. Bongers et al. (2021) give cyclic SCMs their solution-function semantics; Mooij et al. (2013) and Rubenstein et al. (2017) characterize when an

SCM is a consistent abstraction of an underlying dynamical system at equilibrium; Bongers et al. (2018) treat random dynamical systems; Blom et al. (2019) prove the representational impossibility that our case-3 multiplicity branch operationalizes. Iwasaki and Simon (1994) and Dash (2005) are the older warnings that equilibration changes causal structure. Our contribution relative to this line is the learning-theoretic converse direction: not "when does the SCM abstract the dynamics" but "when do the dynamics validate the SCM's do," with the E-stability equivalence and the sharp gain threshold.

Settable systems. White and Chalak (2009) extend Pearl's causal model explicitly for optimization, equilibrium, and learning, with attributes and partitioned settings — framework-level prior art for everything here. The delta is theorem-level: settable systems provide the language for interventions on systems with equilibria; we provide identification conditions, rates, thresholds, and certificates for when the equilibrium answer is the learning answer.

Adaptive learning in macroeconomics. E-stability and least-squares learning convergence are due to Marcet and Sargent (1989) and Evans and Honkapohja (2001); Bullard and Mitra (2002) apply them to monetary policy rules; Evans and Ramey (2006) connect learning to the Lucas critique. This literature has the stability theory but no causal semantics; the cyclic-SCM literature has the semantics but no stability theory. T1 is the bridge, and the certificate is its operational form.

Learning in games and econometrics of learning agents. No-regret convergence to coarse correlated equilibria is classical (Foster and Vohra, 1997; Hart and Mas-Colell, 2000; Cesa-Bianchi and Lugosi, 2006; Fudenberg and Levine, 1998). Nekipelov et al. (2015) use no-regret as an identifying assumption; Bergemann et al. (2022), Magnolfi and Roncoroni (2023), and Syrgkanis et al. (2017) compute LP identified sets over equilibrium polytopes; Hammond et al. (2023) and Mishra and Fox (2024) define interventions over equilibrium sets in causal games. Section 4 states our precise delta against each. Non-convergence of learning dynamics is likewise classical (Sato et al., 2002; Mertikopoulos et al., 2018; Pangallo et al., 2019); we use it as the adversarial test bed for the BOUNDED rung rather than as a subject in itself.

Performative prediction. The single-learner cousin of our question: a predictor whose deployment shifts the distribution it predicts, with stability conditions for convergence to performatively stable points (Perdomo et al., 2020). The trichotomy can be read as the multi-agent, causal-semantics generalization: performative stability is the analogue of case 1, and our case 3 diagnostics have no analogue there because the single-learner setting suppresses equilibrium selection.

8 Limitations and open problems

Exactness class. T1 is exact for linear cyclic SCMs; T1$'$ is local and requires hyperbolicity; the global nonlinear statement leans on Assumption 5.1, parts of which ((R2)–(R3)) are guarded by runtime diagnostics rather than decided a priori. We view the typed degradation to EMPIRICAL as the honest response, but a sharper decidable characterization for structured nonlinear classes (e.g. monotone or gradient-like mean fields) is open. **Boundary cases.** margin $= 0$ and singular $I - B_{\mathrm{do}}$ are refused, not classified; a bifurcation-aware treatment is open. **Game scale.** The CCE LP is exponential in the number of players; sampled and relaxed bounds with certified gaps are the scale path. **Stage-game scope.** Remark 4.6: the demarcation of which learning families escape the stage-game polytope at long horizons — and the reading of the measured ε_T as an enforcement diagnostic — is developed in companion work, with the robustness battery that claim genuinely needs. **Economics-facing model.** E4 is a bespoke static-timing toy; its threshold is Taylor-principle-*type* only. A standard-timing calibrated New-Keynesian learning exercise is required before policy conclusions, and is deferred. **Multiplicity diagnostics.** E7's ensemble basin masses are computed by the experiment, not yet by a shipped instrument; promoting the multiplicity-aware hedge into the certificate layer (all stable σ-solutions plus basin-mass estimates) is engineering follow-up. **E2's dynamical label.** We verified non-convergence and containment, not a positive Lyapunov exponent for our exact configuration; the diagnostic is cheap and planned.

9 Conclusion

"When can you trust an equilibrium counterfactual?" has, for well-delimited classes, a three-part answer that a machine can check: *trust it* exactly when the mean dynamics are E-stable and the stable solution is globally unique — E-stability *is* causal validity of the σ-solution; *bound it* by the measured-regret CCE interval when the system is a finite game, with no assumptions beyond the trajectory log; *and otherwise do not* — but demand the diagnostics, because certified failure (a negative margin no gain can rescue, an abstention in place of a vacuous bound, a basin-mass mixture in place of a fabricated point) is itself usable knowledge. The Lucas critique, in this reading, is not a slogan about the fragility of equilibrium reasoning but a computable property of an intervened system — and the experiments show it firing, quantitatively, exactly where it should.

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A Proof of Theorem 3.2 (stochastic parts)

Throughout, $e_k = x_k - x^*$ and $\nu_1, \ldots, \nu_n$ are the eigenvalues of $A := B_{\mathrm{do}} - I$ (so margin > 0 iff A is Hurwitz).

(2), noise bound. With constant gain γ and $u_{\mathrm{do},k} = \mathbb{E}[u_{\mathrm{do}}] + \xi_k$, where $\{\xi_k\}$ is a martingale-difference sequence with $\sup_k \mathbb{E}[\|\xi_k\|^2 \mid \mathcal{F}_{k-1}] \le \sigma^2$, the error recursion is

$$e_{k+1} = M_\gamma e_k + \gamma \xi_k, \qquad M_\gamma = I + \gamma A.$$

Let $\Sigma_k = \mathbb{E}[e_k e_k^\top]$. Then $\Sigma_{k+1} = M_\gamma \Sigma_k M_\gamma^\top + \gamma^2 \mathbb{E}[\xi_k \xi_k^\top]$, and for $\gamma < \gamma^*$ we have $\rho(M_\gamma) < 1$, so Σ_k converges to the unique solution $\Sigma_\infty(\gamma)$ of the discrete Lyapunov equation $\Sigma = M_\gamma \Sigma M_\gamma^\top + \gamma^2 \Sigma_\xi$ with $\|\Sigma_\xi\| \le \sigma^2$, namely $\Sigma_\infty(\gamma) = \sum_{j \ge 0} \gamma^2 M_\gamma^j \Sigma_\xi (M_\gamma^\top)^j$. Choose a norm in which $\|M_\gamma\| \le 1 - c\gamma$ for some $c > 0$ depending only on the spectrum of A (possible for all $\gamma \le \bar{\gamma} < \gamma^*$ since $\rho(M_\gamma)^2 \le 1 - 2\gamma\, c_0 + \gamma^2 c_1$ with $c_0 = \min_i(-\operatorname{Re} \nu_i) > 0$). Then $\|\Sigma_\infty(\gamma)\| \le \gamma^2 \sigma^2 \sum_j (1 - c\gamma)^{2j} \le \gamma^2 \sigma^2 / (2c\gamma - c^2\gamma^2) = O(\gamma)$, which is the claim $\limsup_k \mathbb{E}\|e_k\|^2 = O(\gamma)$.

(1) ⇒ (3). For decreasing gains satisfying $\sum \gamma_k = \infty$, $\sum \gamma_k^2 < \infty$, the recursion (3) is a Robbins–Monro scheme whose mean ODE is $\dot{e} = Ae$. For linear A Hurwitz, x^* is the globally asymptotically stable equilibrium of the mean ODE; the iterates are bounded almost surely (the quadratic Lyapunov function $V(e) = e^\top P e$ with $A^\top P + PA = -I$ yields a supermartingale argument standard for linear SA; see Kushner and Yin, 2003, Ch. 5 or Borkar, 2008, Ch. 3), and the ODE-method limit-set theorem (Ljung, 1977; Benaïm, 1999; Kushner and Yin, 2003) gives $x_k \to x^*$ a.s.

(3) ⇒ (1) and the converse claim. Suppose margin < 0, i.e. some $\operatorname{Re} \nu_i > 0$; by hyperbolicity we may assume no eigenvalue has zero real part (the boundary case is refused by the instruments, Table 1). Then x^* is a linearly unstable zero of the mean field. Under the excitation condition (A4) of Appendix B (nondegenerate noise in the unstable directions) and gains $\gamma_k = \Theta(k^{-\alpha})$, $\alpha \in (1/2, 1]$, the nonconvergence theorem of Pemantle (1990) (repeller case) and Brandière and Duflo (1996) (saddle/"trap" case) yields $\mathbb{P}(x_k \to x^*) = 0$. Hence (3) fails, and contrapositively (3) ⇒ (1). For the constant-gain converse, Lemma 3.1 gives $|1 + \gamma \nu_i| > 1$ for every $\gamma > 0$, so the mean iteration diverges for every initial condition with a nonzero component in the corresponding invariant subspace — all but a proper (measure-zero) subspace. □

B Assumptions and proof of Theorem 3.4

Assumptions.

(A1) $f \in C^1$ on an open set containing x^*; $F = f - \mathrm{id}$; x^* is a hyperbolic zero of F: $J = DF(x^*)$ has no eigenvalue on the imaginary axis. DF is locally Lipschitz at x^*.

(A2) $x_{k+1} = x_k + \gamma_{k+1}(F(x_k) + M_{k+1})$ with $\{M_k\}$ a martingale-difference sequence with respect to a filtration $\{\mathcal{F}_k\}$: $\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = 0$ and $\mathbb{E}[\|M_{k+1}\|^2 \mid \mathcal{F}_k] \le \sigma^2(1 + \|x_k\|^2)$ a.s.

(A3) Gains: either constant $\gamma_k \equiv \gamma$, or decreasing with $\sum_k \gamma_k = \infty$ and $\sum_k \gamma_k^2 < \infty$.

(A4) (For part (ii) only.) There exist $b, \delta > 0$ such that on $\{\|x_k - x^*\| \le \delta\}$, $\mathbb{E}[\langle M_{k+1}, v\rangle^+ \mid \mathcal{F}_k] \ge b$ for every unit vector v, the noise is a.s. bounded on that event, and $\gamma_k = \Theta(k^{-\alpha})$ with $\alpha \in (1/2, 1]$.

Part (i), constant gain (deterministic core). The damped map is $T_\gamma(x) = x + \gamma F(x)$ with $DT_\gamma(x^*) = I + \gamma J$. By Lemma 3.1 applied to the eigenvalues of J, $\rho(I + \gamma J) < 1$ for every $\gamma \in (0, \gamma^*(J))$. Pick a norm with $\|I + \gamma J\| \le 1 - c\gamma < 1$; by (A1) there is a ball $\mathcal{B}$ around x^* on which $\|T_\gamma(x) - x^*\| \le (1 - c\gamma/2)\|x - x^*\|$, giving local exponential stability with basin at least $\mathcal{B}$. With noise, the argument of Appendix A applied to the linearization, plus a standard perturbation bound for the locally Lipschitz remainder (Benveniste et al., 1990, Ch. 9), yields $\limsup_k \mathbb{E}[\|x_k - x^*\|^2 \mathbf{1}_{\{x_j \in \mathcal{B}\ \forall j \le k\}}] = O(\gamma)$: iterates that remain in the basin concentrate at the mean-square $O(\gamma)$ scale.

Part (i), decreasing gain. J Hurwitz makes x^* a (locally) asymptotically stable equilibrium of $\dot{x} = F(x)$ with domain of attraction $\mathcal{D}$. Let $K \subset \mathcal{D}$ be compact. On the event $\Omega_K = \{x_k \in K$ for all large $k\}$, the SA limit-set theorem (Kushner and Yin, 2003, Thm. 5.2.1; Benaïm, 1999; Borkar, 2008, Ch. 2) applies: under (A1)–(A3) the interpolated iterate trajectory is almost surely an asymptotic pseudotrajectory of the mean flow, whose limit set on Ω_K is a compact, invariant, internally chain-transitive subset of K; since the only such subset of a compact set contained in the basin of a hyperbolic attractor is $\{x^*\}$ itself, $x_k \to x^*$ a.s. on Ω_K. This is the precise sense of "locally, on the basin of x^*": the theorem conditions on remaining in a compact subset of the basin, and Section 6.4 is the measured demonstration of why the conditioning cannot be dropped when multiple attractors exist.

Part (ii). x^* hyperbolic with an unstable eigenvalue is a repeller or a saddle of the mean flow. Under (A2)–(A4), Pemantle (1990) (repellers) and Brandière and Duflo (1996) (saddles; see also Benaïm, 1999) give $\mathbb{P}(x_k \to x^*) = 0$. The excitation condition (A4) is what rules out the degenerate case of noise supported on the stable manifold; it is satisfied, e.g., by any noise with a density bounded below on a neighborhood of the origin, and in particular by the Gaussian noise used in Section 6.4. □

C Proofs for Section 4

Corollary 4.2. The feasible set $\mathrm{CCE}_\varepsilon(G_{\mathrm{do}})$ is a polytope $\{\mu \ge 0,\ \mathbf{1}^\top \mu = 1,\ G\mu \le \varepsilon\}$ whose constraint matrix is fixed. It is nonempty for every $\varepsilon \ge 0$: any Nash equilibrium of the finite game G_{do} — which exists — lies in CCE_0, and the set only grows with ε. The optimal value of a bounded feasible LP is a piecewise-linear, hence locally Lipschitz, function of the constraint right-hand side, with one-sided derivatives given by the optimal dual multipliers (standard LP sensitivity). If $\varepsilon_T \to 0$, any weak limit μ_∞ of $\{\mu_T\}$ satisfies $G\mu_\infty \le 0$ by continuity of $\mu \mapsto G\mu$ on the simplex, so $\mu_\infty \in \mathrm{CCE}_0(G_{\mathrm{do}})$, and the LP values at ε_T converge to the values at 0 by the Lipschitz property. □

Corollary 4.3. If h is constant over CCE_0, the interval is a point and contains the limit of $\mathbb{E}_{\mu_T}[h]$ for any no-regret population by Corollary 4.2. Conversely, if h is non-constant over CCE_0, pick $\mu \in \mathrm{CCE}_0$ with $\mathbb{E}_\mu[h]$ different from the equilibrium point prediction; a population playing i.i.d. draws from μ is no-regret (its realized regrets converge to the nonpositive expected deviation gains) and its time-average converges to $\mathbb{E}_\mu[h]$. Zero-sum example: for any $\mu \in \mathrm{CCE}_0$ of a two-player zero-sum game, the no-deviation constraints give $\mathbb{E}_\mu[u_1] \ge \max_{a_1'} \mathbb{E}_\mu[u_1(a_1', s_2)] \ge v$ and symmetrically $\mathbb{E}_\mu[u_1] \le v$, so every CCE achieves the value v. □

Corollary 4.4. Parametric LP: the max-side value $V(\varepsilon) = \max\{c^\top\mu : \mu \in \mathrm{CCE}_\varepsilon\}$ is concave and piecewise-linear in the constraint right-hand side (standard RHS-perturbation concavity of a maximization LP), and nondecreasing because relaxing constraints can only increase the value. Along the uniform inflation direction $\varepsilon = \varepsilon_0 + t\mathbf{1}$, at any t where the optimal dual is unique the derivative is $\sum_j y_j^*$, the sum of the optimal dual multipliers on the inflated constraints; at kinks the one-sided derivatives are the extreme values of that sum over the optimal dual face. The min side is the max side of $-h$, hence convex with the symmetric statement. □

Remark C.1. Our LP solver returns the dual vector alongside the primal optimum, and the shipped certificates carry $\{\partial\text{lower}, \partial\text{upper}, \partial\text{width}\}$ at the measured ε_T. All experimental LP values in this paper are additionally verified by independent weak-duality certificates: a dual-feasible $y \geq 0$ with $(G^\top y)_j \geq v - c_j$ for all j certifies that no feasible μ exceeds v, with slack $\sim 10^{-14}$ in all reported cases.

D Proof details for Theorem 5.2

Exhaustiveness. Fix the class. (a)/(b): by (R1) the intervened mean field has finitely many zeros, each hyperbolic, so each is either stable (margin > 0) or unstable (margin < 0); the number of stable zeros is 0, 1, or ≥ 2. If (R2)/(R3) fail at runtime (divergence, cycles, positive Lyapunov estimate), case 3 fires by its second clause. (c): the LP width is a number; it is either $\leq$ tol (case 1(ii)) or $>$ tol, and in the latter case the measured-ε_T interval is either informative (case 2) or vacuous (case 3). Every system in the stated classes therefore lands in some case.

Exclusivity. Case 1(i) requires a globally unique stable σ-solution, which the multiplicity clause of case 3 negates; case 1(ii) and case 2 split on width $\leq$ tol versus $>$ tol; the remaining clauses of case 3 (all-unstable, (R)-failure, vacuity) each contradict the preconditions of cases 1 and 2 directly.

Warrant. Case 1(i): in class (a), Theorem 3.2 gives global convergence for every $\gamma \in (0, \gamma^*)$ and a.s. convergence under decreasing gain — globality is free because the mean dynamics are linear. In class (b), let x^* be the unique stable zero. By (R2) the iterates a.s. remain in a compact set K; by the limit-set theorem the a.s. limit set is an internally chain-transitive subset of the flow restricted to K, which by (R3) is a subset of the zero set. By Theorem 3.4(ii) (whose excitation condition we adopt here) the unstable zeros attract with probability zero, so the limit set is a.s. $\{x^*\}$. Case 1(ii) is Corollary 4.3; case 2 is Theorem 4.1 plus Corollary 4.4; case 3's clauses carry, respectively, the divergence certificate of Theorem 3.2's converse, the runtime diagnostics, the abstention rule of Remark 4.5, and the basin-local certificates plus the representational impossibility of Blom et al. (2019) for the multiplicity clause.

Decidability scope. In class (a) every condition is a spectral or LP computation. In class (b) the theorem is decidable *given* the enumeration of zeros (available by bisection for the scalar systems used here, and by homotopy or interval methods more generally); (R2)–(R3) are guarded, not decided — the honest scope stated in Assumption 5.1. In class (c) both conditions are single LP solves. □

E Experimental details and reproducibility

All experiments are single-file scripts against the released instruments; seeds are fixed in-script; every number in Section 6 is pasted from an actual run. LP values carry independent weak-duality verification (Appendix C). In the artifact's API (library name withheld here; replaced by an anonymized artifact link at submission), the instruments are: the comparator `compare_equilibrium_unrolling`; the spectral diagnostics `stability_margin` and `max_stable_learning_rate`; and the game instruments `cce_polytope`, `cce_bounds`, `cce_regret`, and `certify_cce_do`.

E1 (cobweb). Demand $D(p) = 10 - p$, supply intercept $c = 1$, tax do(τ=1) entering as the supply-intercept shift; SCM $p = -(s/d)\, p^e + (a - c + s\tau)/d$, $p^e = p$. Regimes: A $s/d = 0.5$; B $s/d = 1.5$ (naive, then damped $\gamma = 0.4 = \gamma^*/2$); C trend-extrapolating demand $b = 1.2$; D arctan supply (scale 2.2, steepness 12) under adaptive expectations $p^{e\prime} = (1-g)p^e + g\, f(p^e)$, with the chaotic gain $g = 0.14$ found by a Lyapunov scan over $g \in [0.08, 0.52]$ and the largest Lyapunov exponent +0.222 computed over 8,000 steps. The linearized certificate at p^* hedges at $\gamma^* = 0.073 < g$.

E2 (perturbed RPS). Payoffs as in Sato et al. (2002) with $\epsilon_X = 0.5$, $\epsilon_Y = -0.1$; intervention: payoff tax 0.3 on X's action 0 (the intervened game built explicitly). Hedge (multiplicative-weights) population, T = 200,000 rounds, seed 7. Exact CCE interval for X's payoff $[0.0667, 0.3091]$; measured-ε_T interval $[0.0646, 0.3120]$ at $\varepsilon_T = 0.0021$.

E4 (macro loop). $\beta = 0.99$, $\sigma = 1$, $\kappa = 0.3$, demand shock do(u=+1); $T(\varphi) = (\beta + \kappa\sigma)/(1 + \kappa\sigma\varphi)$; cyclic SCM $B = [[0, T], [1, 0]]$ closed by naive expectations; comparator horizon 2,000, tolerance 10^{-6}; learning paths are the naive-expectations recursion (thirty steps shown). $\varphi = 1.5$: margin = +0.057, gap 7×10^{-15}, $\pi^* = +9.06$, $\pi_{30} = +8.79$. $\varphi = 0.5$: margin = −0.059, $\gamma^* = 0$, $\pi^* = -8.21$, $\pi_{30} = +249.6$.

E7 (bistable selection). $F(x) = \tanh(3x) - x + u$; roots by bisection on a grid of 6×10^5 points over $[-3, 3]$; margins = $-F'$(root). SA ensemble: $N = 10^5$ learners, constant gain 0.05, additive Gaussian noise (s.d. 0.1), 4,000 steps, $x_0 \sim \mathcal{N}(0, s_0^2)$, seed 0; dispersion curve over $s_0 \in \{0.25, 0.5, 1.0, 1.5, 2.0, 3.0\}$. Figures regenerated deterministically by the released figure script from the same code paths and seeds.